

US Patent & Trademark Office

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United States Patent Application Publication

20250273005

Kind Code

A1

Publication Date

August 28, 2025

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FINGERPRINT SENSOR PACKAGES AND DEVICES INCLUDING THE SAME

Abstract

A fingerprint sensor package, comprising: a package substrate, wherein an upper surface of the package substrate includes a sensing region and a peripheral region that extends around the sensing region, and wherein a lower surface of the package substrate includes a mounting region and a connection region that extends around the mounting region; external connection pads in the connection region on the lower surface of the package substrate; a lower protection layer on the lower surface of the package substrate, wherein the lower protection layer includes openings that overlap at least a portion of the external connection pads respectively, wherein the lower protection layer includes a protection layer pattern that overlaps the external connection pads in the openings; and an anisotropic conductive layer on the external connection pads in the openings and on the lower protection layer.

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Family ID: 1000008245210

Appl. No.: 18/922517

Filed: October 22, 2024

Foreign Application Priority Data

KR 10-2024-0026518

Feb. 23, 2024

Publication Classification

Int. Cl.: G06V40/13 (20220101); H01L21/56 (20060101); H01L23/00 (20060101); H01L23/31 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority benefit of under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0026518, filed on Feb. 23, 2024 in the Korean Intellectual Property Office (KIPO), the disclosures of which are herein incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] Example embodiments of the present inventive concepts relate to fingerprint sensors. More particularly, example embodiments of the present inventive concepts relate to capacitive fingerprint sensor packages and devices (e.g., smart cards) including the same.

2. Description of the Related Art

[0003] In order to prevent loss and theft of (payment) devices (e.g., credit cards), the demand for (payment) devices (e.g., credit cards) with a fingerprint authentication function is increasing. In order to manufacture a (payment) device (e.g., a credit card) with the fingerprint authentication function, a fingerprint sensor package capable of fingerprint recognition may be mounted on the device. For example, a payment device (e.g., a credit card), including a sensor (e.g., a fingerprint sensor package) with an authentication function (e.g., a fingerprint authentication function) may be referred to as a smart card. An anisotropic conductive film (ACF) may be used to mount the fingerprint sensor package on the smart card. When a force is applied to compress the anisotropic conductive film, there may be a problem in that the anisotropic conductive film escapes from a region in which a connection pad is disposed by pressure and then contacts a redistribution wiring or a ground pattern located on an upper surface of the fingerprint sensor package, thereby causing a short circuit.

SUMMARY OF THE INVENTION

[0004] Example embodiments may provide a fingerprint sensor package capable of preventing a short circuit due to escape of an anisotropic conductive layer.

[0005] Example embodiments may provide a smart card including the fingerprint sensor package.

[0006] In accordance with example embodiments, a fingerprint sensor package, comprising: a package substrate including an upper surface and a lower surface that is opposite to the upper surface in a first direction, wherein the first direction is perpendicular to the upper surface of the package substrate, wherein the upper surface of the package substrate includes a sensing region and a peripheral region that extends around the sensing region, and wherein the lower surface of the package substrate includes a mounting region and a connection region that extends around the mounting region; external connection pads in the connection region on the lower surface of the package substrate; a plurality of capacitor patterns in the sensing region on the upper surface of the package substrate; a control portion in the mounting region on the lower surface of the package substrate, wherein the control portion includes a sensor chip and a passive element that is adjacent the sensor chip; a lower protection layer on the lower surface of the package substrate, wherein the lower protection layer includes openings that overlap at least a portion of the external connection pads in the first direction respectively, wherein the lower protection layer includes a protection layer pattern that overlaps the external connection pads in the first direction in the openings; and an anisotropic conductive layer on the external connection pads in the openings and on the lower

protection layer.

[0007] In accordance with example embodiments, a fingerprint sensor package, comprising: a package substrate including: a first substrate having a sensing region and an adhesion region extending around the sensing region on an upper surface thereof and having first bonding pads on a lower surface thereof; and a second substrate having second bonding pads and external connection pads on a lower surface thereof, the second bonding pads being electrically connected to the first bonding pads through conductive wires, the second substrate having an opening region that overlaps the sensing region of the upper surface of the first substrate in a first direction, the second substrate being adhered on the adhesion region of the upper surface of the first substrate; a plurality of capacitor patterns in the sensing region on the upper surface of the first substrate; a control portion on the lower surface of the first substrate, the control portion including a sensor chip and a passive element adjacent the sensor chip; a lower protection layer on the lower surface of the second substrate and having openings that overlap the external connection pads in the first direction respectively, the lower protection layer having a protection layer pattern that overlaps a portion of the external connection pads in the openings in the first direction; and an anisotropic conductive layer on the external connection pads in the openings and on the lower protection layer, wherein the first direction is perpendicular to the upper surface of the first substrate.

[0008] In accordance with example embodiments, a fingerprint sensor package, comprising: a package substrate including an upper surface and a lower surface that is opposite to the upper surface in a first direction, wherein the first direction is perpendicular to the upper surface of the package substrate, wherein the upper surface of the package substrate includes a first region and a second region that extends around the first region, and wherein the lower surface of the package substrate includes a third region and a fourth region that extends around the third region; external connection pads in the fourth region on the lower surface of the package substrate; a plurality of first sensing patterns in the first region, extending in a second direction, being spaced apart from each other in a third direction; a plurality of second sensing patterns in the first region, wherein the plurality of second sensing patterns are spaced apart from the plurality of first sensing patterns in the first direction, and wherein the plurality of second sensing patterns extend in the third direction and spaced apart from each other in the second direction; a control portion in the third region, the control portion including a sensor chip and a passive element adjacent the sensor chip; a lower protection layer on the lower surface of the package substrate and having openings that overlap at least a portion of the external connection pads in the first direction respectively, and the lower protection layer including a protection layer pattern that overlaps the external connection pads in the first direction in the openings; and an anisotropic conductive layer on the external connection pads in the openings and on the lower protection layer, wherein the second direction and the third direction are parallel with the upper surface of the package substrate, and wherein the second direction and the third direction intersect with each other.

[0009] In accordance with example embodiments, a fingerprint sensor package may include a lower protection layer covering the lower surface of the package substrate and having a protection pattern having openings exposing the external connection pads respectively, and covering at least a portion of the exposed external connection pads in each of the openings and an anisotropic conductive layer covering the exposed external connection pads in each of the openings and covering the lower protection layer in a form of a band shape.

[0010] The openings and the external connection pads may have a certain step to provide an accommodation space therein. Accordingly, when performing the compression process for mounting the fingerprint sensor package on a smart card, the conductive balls in the anisotropic conductive film may be trapped in the accommodation space without escaping the fingerprint sensor package.

[0011] Thus, short circuits in the fingerprint sensor package due to the escape of the anisotropic conductive film can be prevented, and more conductive balls come into contact with the external

connection pad compared to the prior art, thereby increasing electrical conductivity and improving the stability of bonding.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] Example embodiments of the present inventive concepts will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings.

FIGS. **1** to **21** represent non-limiting, example embodiments as described herein.

[0013] FIGS. **1** to **9** are views illustrating a fingerprint sensor package in accordance with example embodiments.

[0014] FIGS. **10** to **18** are views illustrating a method for manufacturing a smart card in accordance with example embodiments.

[0015] FIGS. **19** to **21** are views illustrating a fingerprint sensor package in accordance with example embodiments.

DETAILED DESCRIPTION OF THE INVENTION

[0016] Hereinafter, example embodiments of the present inventive concepts will be explained in detail with reference to the accompanying drawings.

[0017] FIG. **1** is a plan view illustrating a fingerprint sensor package in accordance with example embodiments. FIG. **2** is a cross-sectional view taken along the line A-A' in FIG. **1**. FIG. **3** is a cross-sectional view taken along the line B-B' in FIG. **1**. FIG. **4** is a bottom view illustrating a package substrate of the fingerprint sensor package of FIG. **1**.

[0018] Referring to FIGS. **1** to **4**, a fingerprint sensor package **10** may include a package substrate **100**, a sensing portion **200** provided on the package substrate **100**, a coating member **300** on (e.g., overlapping or covering) the sensing portion **200**, a control portion **400** including a sensor chip **410** that is configured to process information obtained from the sensing portion **200**, a protection layer **500**, a molding member **600** on (e.g., overlapping or covering) the control portion **400**, and an anisotropic conductive film ACF on (e.g., overlapping or covering) a lower protection layer **520** (in a band shape in a plan view). The anisotropic conductive film ACF may be referred to as an anisotropic conductive layer ACF. In some embodiments, the coating member **300** may overlap (e.g., cover) the sensing portion **200** (in a third direction). The molding member **600** may overlap (e.g., cover) the control portion **400** (in a first direction, a second direction, and/or the third direction). The anisotropic conductive layer ACF may overlap (e.g., cover) the lower protection layer **520** (in the first direction, the second direction, and/or the third direction). As used herein, “an element A overlapping an element B in a direction X” (or similar language) means that there is at least one line that extends in the direction X and intersects both the elements A and B. The first direction and the second direction may be parallel with an upper surface **104** and/or a lower surface **102** of the package substrate **100**. The first direction and the second direction may intersect with (may be perpendicular to) each other. The third direction may be perpendicular to the upper surface **104** and/or the lower surface **102** of the package substrate **100**.

[0019] In example embodiments, when a user contacts the sensing portion **200** (the fingerprint sensor), the fingerprint sensor package **10** may recognize a fingerprint of the user. The fingerprint sensor package **10** may compare the information on the fingerprint with a pre-stored fingerprint information to verify the identity of the user. When the user contacts the sensing portion **200**, different deformation occurs in portions (e.g., each portion) of the sensing portion **200** (the fingerprint sensor) due to the irregularities of the fingerprint. The fingerprint sensor package **10** may include a capacitive fingerprint sensor that determines the shape of the fingerprint by calculating the value of the deformation based on a value difference in capacitance recognized by the sensor. For example, the sensing portion **200** may be the capacitive fingerprint sensor.

[0020] As illustrated in FIGS. 2 and 3, the package substrate **100** may include a printed circuit board (PCB). The package substrate **100** may be a printed circuit board including redistribution wirings of a multi-layered structure. The package substrate **100** may have, for example, a shape of a rectangular plane or a square plane (in a plan view). In some embodiments, the package substrate **100** may have a shape in which each corner is rounded (in a plan view), and thus, cracks may be effectively prevented from occurring at the corners in the process of cutting the package substrate **100** through a punching process. However, the shape of the package substrate **100** is not limited to the above-noted shapes.

[0021] The package substrate **100** may include first, second, and third insulating layers **110**, **120**, and **130** sequentially stacked, a plurality of first and second redistribution wirings **114** and **124** provided in the first, second, and third insulating layers **110**, **120**, and **130**, a plurality of first, second, and third through vias **112**, **122**, and **132**, electrode pads **116**, and external connection pads **118**.

[0022] The package substrate **100** may further include a conductive external connection member on a lower surface **102** of the package substrate **100**. In accordance with a type of the conductive external connection member, the package substrate **100** may include a ball grid array (BGA) or a land grid array (LGA).

[0023] The first, second, and third insulating layers **110**, **120**, and **130** may include a polymer, a dielectric layer, etc. For example, the first, second, and third insulating layers **110**, **120**, and **130** may include a photosensitive insulating layer such as a photo imageable dielectric (PID). The first, second, and third insulating layers **110**, **120**, and **130** may be formed by a vapor deposition process, a spin coating process, etc. The first and second redistribution wirings **114** and **124** and the first, second, and third through vias **112**, **122**, and **132** may include, for example, aluminum (Al), copper (Cu), tin (Sn), nickel (Ni), gold (Au), platinum (Pt), and/or an alloy thereof. The first and second redistribution wirings **114** and **124** may be formed by a plating process, an electroless plating process, a vapor deposition process, etc.

[0024] As illustrated in FIG. 4, a lower surface of the first insulating layer **110**, that may be, the lower surface **102** of the package substrate **100** may include a mounting region MR and a connection region CR extending around (e.g., surrounding) the mounting region MR (in the plan view). The electrode pads **116** may be provided in the mounting region MR on the lower surface of the first insulating layer **110**. The electrode pads **116** may be arranged in an array form in the mounting region MR. The sensor chip **410** and passive elements **420** may be on the lower surface of the first insulating layer **110** in the mounting region MR (mounted in the mounting region MR). The electrode pads **116** may electrically connect the sensor chip **410**, the passive elements **420**, and the package substrate **100**. It will be understood that when an element or layer is referred to as being “on”, “connected to”, “coupled to”, or “responsive to” another element or layer, it may be directly on, directly connected to, directly coupled to, or directly responsive to the other element or layer, or one or more intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on”, “directly connected to”, “directly couple to,” or “directly responsive to”, another element, there are no intervening elements present. In addition, “electrical connection” conceptually includes a physical connection and a physical disconnection.

[0025] The external connection pads **118** may be provided on the lower surface of the first insulating layer **110** in the connection region CR. The external connection pads **118** may electrically connect the fingerprint sensor package **10** to a (payment) device (e.g., a smart card) when the fingerprint sensor package **10** is mounted on the (payment) device (e.g., the smart card). The external connection pads **118** may have, for example, a rectangular shape with a long side, a short side and rounded corners (in a plan view). In some embodiments, a length of the long side may be within a range of (about) 2.4 mm to (about) 2.8 mm. A length of the short side may be within a range of (about) 1.3 mm to (about) 1.7 mm. The shape and size of the external connection pads **118** are not limited to above-noted descriptions. The external connection pads **118** may be

disposed to be spaced apart from each other along the periphery of the mounting region MR. For example, the external connection pads **118** may be spaced apart from each other in the first direction and/or the second direction.

[0026] The first through vias **112** may be provided in the first insulating layer **110**. The first through vias **112** may extend in (e.g., penetrate) the first insulating layer **110** (in the third direction). In some embodiments, the first through vias **112** may extend from the lower surface of the first insulating layer **110** to an upper surface opposite to the lower surface of the first insulating layer **110** (in the third direction). The first through vias **112** may be electrically connected to the electrode pads **116** in the mounting region MR and the external connection pads **118** in the connection region CR.

[0027] The first redistribution wirings **114** may be provided on the first insulating layer **110**. The first redistribution wirings **114** may be electrically connected to the electrode pads **116** and the external connection pads **118** through the first through vias **112** in the first insulating layer **110**.

[0028] The second insulating layer **120** may be provided on the first insulating layer **110**. The second through vias **122** may be provided in the second insulating layer **120**. The second through vias **122** may extend in (e.g., penetrate) the second insulating layer **120** (in the third direction). In some embodiments, the second through vias **122** may extend from a lower surface of the second insulating layer **120** to an upper surface of the second insulating layer **120** opposite to the lower surface (in the third direction). In some embodiments, the second through vias **122** may be (electrically) connected to the first redistribution wirings **114** provided on the first insulating layer **110**.

[0029] The second redistribution wirings **124** may be provided on the second insulating layer **120**. The second redistribution wirings **124** may be connected (e.g., electrically connected) to the first redistribution wirings **114** through second through vias **122** in the second insulating layer **120**.

[0030] The third insulating layer **130** may be provided on the second insulating layer **120**. The third through vias **132** may be provided in the third insulating layer **130**. The third through vias **132** may extend in (e.g., penetrate) the third insulating layer **130** (in the third direction). In some embodiments, the third through vias **132** may extend from a lower surface of the third insulating layer **130** to an upper surface of the third insulating layer **130** opposite to the lower surface (in the third direction). In some embodiments, the third through vias **132** may be (electrically) connected to the second redistribution wirings **124** provided on the second insulating layer **120**.

[0031] The upper surface of the third insulating layer **130**, that may be, the upper surface **104** of the package substrate **100** may include a sensing region SR and a peripheral region PR extending around (e.g., surrounding) the sensing region SR (in a plan view). The sensing region SR may be a region in which the sensing portion **200** including a first sensing pattern **210** and a second sensing pattern **220** for fingerprint recognition is disposed. The sensing region SR may be located in a central region of the third insulating layer **130** (in a plan view), but is not limited thereto. The sensing region SR may have a rectangular shape in a plan view, but is not limited thereto.

[0032] A ground pattern **140** may be provided in the peripheral region PR on the upper surface of the third insulating layer **130**. The ground pattern **140** may provide a reference potential (e.g., reference electric potential or reference electrostatic potential) to the sensor chip **410** and may serve to shield noise when the sensing portion **200** senses a user's fingerprint. The ground pattern **140** may have an annular shape to extend around (e.g., surround) the sensing region SR (in a plan view). For example, the ground pattern **140** may be disposed horizontally along the periphery of the sensing region SR and may be formed in a closed annular pattern having a predetermined width. For example, the sensing region SR may be between portions of the ground pattern **140** in a cross-sectional view. The ground pattern **140** may be electrically connected to the sensor chip **410** through the first, second, and third through vias **112**, **122**, and **132** and the first and second redistribution wirings **114** and **124**.

[0033] The sensing portion **200** may include a plurality of first sensing patterns **210** and a plurality

of second sensing patterns **220** as a plurality of capacitor patterns. The first sensing patterns **210** may be provided on the upper surface of the third insulating layer **130**. In some embodiments, each of the first sensing patterns **210** may have a linear shape extending in the first direction on the upper surface of the third insulating layer **130**. The first sensing patterns **210** may be disposed to be spaced apart from each other at regular intervals in the second direction. At least one end of both ends of the first sensing patterns **210** extending in the first direction may be (electrically) connected to the third through vias **132**, and each of the neighboring first sensing patterns **210** may be electrically separated (e.g., insulated) from each other.

[0034] The second sensing patterns **220** may be provided to be spaced apart from the first sensing patterns **210** on the upper surface of the third insulating layer **130** in the third direction. Each of the second sensing patterns **220** may have a linear shape extending in the second direction. The second sensing patterns **220** may be disposed to be spaced apart from each other in the first direction. At least one end of both ends of the second sensing patterns **220** extending in the second direction may be (electrically) connected to the third through vias **132**, and each of the neighboring second sensing patterns **220** may be electrically separated (e.g., insulated) from each other.

[0035] As illustrated in FIG. 1, when viewed in a plan view, the first sensing patterns **210** and the second sensing patterns **220** may be arranged to have a plurality of crossing points to constitute a plurality of sensing pixels. The sensing pixels may be arranged in the first direction and the second direction. That is, the sensing pixels may be arranged in an array form. Since the first sensing patterns **210** and the second sensing patterns **220** are spaced apart in the third direction, the sensing pixels may have capacitance values. A value of capacitance corresponding to each of the sensing pixels may be changed by the capacitance induced by the unevenness of the user's fingerprint. Accordingly, the user's fingerprint information may be recognized by the sensing pixels to identify the user's fingerprint. In the capacitance value, sensing noise may occur in accordance with the shape of the user's fingerprint and the surrounding environment (e.g., the external environment). Accordingly, (in a plan view) the ground pattern **140** may be provided to extend around (e.g., surround) the sensing region SR in order to minimize the sensing noise. As used hereinafter, the terms "external/outside configuration", "external/outside device", "external/outside power", "external/outside signal", or "outside" are intended to broadly refer to a device, circuit, block, module, power, and/or signal that resides externally (e.g., outside of a functional or physical boundary) with respect to a given circuit, block, module, system, or device.

[0036] The coating member **300** may be on (e.g., overlap with in the third direction or cover) the sensing portion **200** including the first sensing patterns **210** and the second sensing patterns **220** in the sensing region SR on the upper surface **104** of the package substrate **100**. The coating member **300** may protect the sensing portion **200** from contamination or damage caused by an external impact on the sensing region SR. The coating member **300** may be provided to directly contact a fingerprint of a user. The coating member **300** may include a material having a dielectric constant suitable for fingerprint recognition. The coating member **300** may protrude from (the upper surface **104** of) the package substrate **100** in the third direction and may be formed to have a thickness of (about) 50 μm or less in the third direction in consideration of sensing sensitivity. For example, the thickness of the coating member **300** may be within a range of (about) 5 μm to (about) 50 μm .

[0037] The control portion **400** may include the sensor chip **410** and the passive element **420**. The control portion **400** may be disposed in the mounting region MR on the lower surface of the first insulating layer **110**, that may be, the lower surface **102** of the package substrate **100**.

[0038] The sensor chip **410** may be configured to perform an operation for recognizing a user's fingerprint based on changes in capacitance values of the sensing pixels. For example, the sensor chip **410** may include a memory chip and/or a processor chip. The passive element **420** may include a capacitor that stores capacitance induced by the first sensing pattern **210** and the second sensing pattern **220**. The sensor chip **410** may be mounted on the package substrate **100** through the electrode pads **116** provided on the lower surface of the first insulating layer **110**. The sensor chip

410 may be electrically connected to the first sensing pattern **210** and the second sensing pattern **220** through the plurality of first and second redistribution wirings **114** and **124** and the plurality of first, second, and third through vias **112**, **122**, and **132**.

[0039] The passive element **420** may include a first conductive terminal **422**, a second conductive terminal **424**, and an insulator **426**. The first conductive terminal **422** and the second conductive terminal **424** may be a first electrode and a second electrode, respectively. The insulator **426** may be provided between the first conductive terminal **422** and the second conductive terminal **424**. A plurality of passive elements **420** may be provided adjacent to the sensor chip **410**. Each of the passive elements **420** may be disposed in parallel to be spaced apart from each other. The passive elements **420** may be mounted on the package substrate **100** through the electrode pads **116** provided on the lower surface of the first insulating layer **110**.

[0040] The protection layer **500** may include an upper protection layer **510** and a lower protection layer **520**. The upper protection layer **510** may be on (e.g., overlap in the third direction or cover) the upper surface **104** of the package substrate **100**. The upper protection layer **510** may be provided on the third insulation layer **130** to expose (at least a portion of) the ground pattern **140**.

[0041] The lower protection layer **520** may be on (e.g., overlap in the third direction or cover) the lower surface **102** of the package substrate **100**. The lower protection layer **520** may include openings **530** that expose (at least a portion of) the external connection pads **118** provided on the lower surface **102** of the package substrate **100**, respectively. Herein, an opening exposing an element A may refer to an opening disposed in an overlapping position with the element A (in the third direction). Each of the openings **530** may have a shape corresponding to a shape of each of the external connection pads **118**. For example, each of the openings **530** may have a size greater than a size of each of the external connection pads **118** (in a plan view) in order to expose (at least a portion of) the external connection pads **118**. For example, when each of the external connection pads **118** has a rectangular shape with rounded corners, each of the openings **530** may have a long side and a short side and may be provided to have a rectangular shape greater than the rectangular shape of the external connection pads **118** in a plan view to expose (at least a portion of) the external connection pads **118**, respectively. A length of the long side of the opening **530** may be within a range of (about) 2.8 mm to (about) 3.2 mm. A length of the short side of the opening **530** may be within a range of (about) 1.7 mm to (about) 2.1 mm. The shapes and sizes of the external connection pads **118** and the openings **530** are not limited to the above-noted descriptions.

[0042] The lower protection layer **520** may expose the electrode pads **116** in the mounting region MR. Since the electrode pads **116** are arranged in an array form in the mounting region MR on the lower surface **102** of the package substrate **100**, the lower protection layer **520** may have openings for exposing electrode pads **116** arranged in an array form similarly to expose the electrode pads **116**. Accordingly, the sensor chip **410** and the passive elements **420** may be electrically connected to the electrode pads **116**, such that the sensor chip **410** and the passive elements **420** may be electrically connected to the sensing portion **200** including the first sensing patterns **210** and the second sensing patterns **220**, that may be, a plurality of capacitor patterns (e.g., sensing pixels), to receive information.

[0043] As illustrated in FIG. 4, the anisotropic conductive layer ACF may be on (e.g., overlap in the third direction or cover) the lower protection layer **520** on the lower surface **102** of the package substrate **100**. The anisotropic conductive layer ACF may have, for example, a band shape (in a plan view), that may be, an annular shape (in the plan view). The anisotropic conductive layer ACF may be provided in the connection region CR extending around (e.g., surrounding) the mounting region MR (in a plan view) on the lower surface **102** of the package substrate **100**. The anisotropic conductive layer ACF may be disposed to overlap in the third direction (e.g., cover) the openings **530** and accommodation grooves **550** on the lower protection layer **520**.

[0044] FIGS. 5 to 8 are enlarged views of the portion 'X' in FIG. 4. FIG. 5 is a plan view illustrating a protection layer pattern **540**. FIG. 6A is a cross-sectional view of the protection layer

pattern **540** taken along the line C-C' in FIG. 5. FIG. 6B is a cross-sectional view of the protection layer pattern **540** taken along the line D-D' in FIG. 5. FIGS. 7 and 8 illustrate plan views of the protection layer pattern **540** according to some embodiments.

[0045] Referring to FIGS. 5 to 8, the lower protection layer **520** may further include a protection layer pattern **540** that overlaps at least a portion of the external connection pads **118** in the opening **530**, when viewed in a plan view. For example, the protection layer pattern **540** may overlap with the external connection pads **118** in the third direction. The protection layer pattern **540** may include a material corresponding to that of the lower protection layer **520**. For example, the lower protection layer **520** and the protection layer pattern **540** may have the same material. In some embodiments, the protection layer pattern **540** may be a portion of the lower protection layer **5320**.

[0046] As illustrated in FIG. 5, the protection layer pattern **540** may include first protrusion portions **542a** and second protrusion portions **542b**. The first protrusion portions **542a** may extend from a first side portion **S1** of the opening **530** in a direction perpendicular to the extending direction of the first side portion **S1** (e.g., in the second direction). The second protrusion portions **542b** may extend from a third side portion **S3** opposite to the first side portion **S1** of the opening **530** in a direction perpendicular to the extending direction of the third side portion **S3** (e.g., in the second direction). In some embodiments, the first protrusion portions **542a** may obliquely extend with a preset angle with respect to the extending direction of the first side portion **S1** of the opening **530**. The second protrusion portions **542b** may obliquely extend with a preset angle with respect to the extending direction of the third side portion **S3** of the opening **530**. Herein, a side portion (e.g., the first side portion **S1**) of the opening **530** may refer to a corresponding side surface of the lower protection layer **520** exposed by the opening **530**.

[0047] The first protrusion portions **542a** may be spaced apart from each other along the first side portion **S1** of the opening **530**. The second protrusion portions **542b** may be spaced apart from each other along the third side portion **S3** of the opening **530**. For example, as illustrated in FIG. 5, three first protrusion portions **542a** and three second protrusion portions **542b** may be arranged along the first side portion **S1** and the third side portion **S3** of the opening **530**, respectively. Each of the first and second protrusion portions **542a** and **542b** may have a width **L1** (in the first direction).

Adjacent ones of the first protrusion portions **542a** may be spaced apart from each other by a spacing distance **L2** (in the first direction). Adjacent ones of the second protrusion portions **542b** may be spaced apart from each other by the spacing distance **L2** (in the first direction). The width **L1** may be at least 80 μm . The spacing distance **L2** between the protrusion portions may be at least 100 μm .

[0048] As illustrated in FIGS. 6A and 6B, the openings **530** may have a step portion **H** of a predetermined height with respect to the external connection pads **118**. The height of the step portion **H** may be a distance from an upper surface of the external connection pads **118** in the third direction. The height of the step portion **H** may be within a range of (about) 15 μm to (about) 25 μm . Accordingly, the openings **530** may provide an accommodation space **A** therein. In order to position (e.g., mount) the fingerprint sensor package **10** on a (payment) device (e.g., a smart card), when the anisotropic conductive layer **ACF** as a conductive connection member is positioned on the external connection pads **118** and then compressed, the anisotropic conductive layer **ACF** may be confined in the accommodation space **A** without escaping the connection region **CR**.

[0049] As illustrated in FIGS. 7 and 8, the protection layer pattern **540** may include first extension portions **544a** and second extension portions **544b**. The first extension portions **544a** may be provided to extend between (from) the first side portion **S1** of the opening **530** and (to) the third side portion **S3** facing the first side portion **S1** (in the second direction). A plurality of first extension portions **544a** may be disposed in parallel to be spaced apart from each other along the first side portion **S1** of the opening **530**. For example, the first extension portions **544a** may be spaced apart from each other in the first direction. The second extension portions **544b** may be provided to extend between (from) a second side portion **S2** perpendicular to the first side portion

S1 of the opening 530 and (to) a fourth side portion S4 facing the second side portion S2 (in the first direction). A plurality of second extension portions 544b may be disposed in parallel to be spaced apart from each other along the second side portion S2 of the opening 530. For example, the second extension portions 544b may be spaced apart from each other in the second direction. The first extension portions 544a may have a width L3 in the extending direction of the first side portion S1 of the opening 530. For example, each of the first extension portions 544a may have a width L3 in the first direction. The second extension portions 544b may have a width L4 in the extending direction of the second side portion S2 of the opening 530. For example, each of the second extension portions 544b may have a width L4 in the second direction. A width (e.g., the width L3) of the first extension portion 544a in the extending direction of the first side portion S1 (e.g., the first direction) and a width (e.g., the width L4) of the second extension portion 544b in the extending direction of the second side portion S2 (e.g., the second direction) may be at least 80 μm , respectively. In some embodiments, the first extension portions 544a and the second extension portions 544b may form an integrated unitary structure. An integrated unitary structure herein may refer to a structure without a visible boundary between sub-elements (e.g., the first extension portions 544a and the second extension portions 544b) thereof.

[0050] The first extension portions 544a and the second extension portions 544b may extend in a direction perpendicular to each other to define a plurality of accommodation cavities 546 that expose the external connection pads 118. Each of the accommodation cavities 546 may have a rectangular or circular shape in a plan view, but not limited thereto. In some embodiments, the width L3 of the first extension portion 544a may be a distance (e.g., the closest distance) between adjacent accommodation cavities 546 arranged in the first direction. The width L4 of the second extension portion 544b may be a distance (e.g., the closest distance) between adjacent accommodation cavities 546 arranged in the second direction. The accommodation cavities 546 may be arranged in an array form. The accommodation cavities 546 may provide an accommodation space A therein. In order to position (e.g., mount) the fingerprint sensor package 10 on a (payment) device (e.g., a smart card), when the anisotropic conductive layer ACF as a conductive connection member is positioned on the external connection pads 118 and then pressed, the anisotropic conductive layer ACF may be confined in the accommodation space A without escaping the connection region CR.

[0051] When the anisotropic conductive layer ACF is placed on the external connection pads 118 and then compressed to mount the fingerprint sensor package 10 on a (payment) device (e.g., a smart card), the protection layer pattern 540 may allow conductive balls CB in the anisotropic conductive layer ACF to be positioned (accommodated) between the protective layer patterns 540 (in the accommodation space of the plurality of accommodation cavities 546), thereby preventing the conductive ball CB from escaping the connection area CR. When viewed in a plan view, the protection layer pattern 540 may (partially) overlap the external connection pad 118 (in the third direction). An area of a portion where the protection layer pattern 540 overlaps the external connection pad 118 may be (about) 20% or less of an area of the external connection pad 118 (in a plan view). Accordingly, an area for stably and electrically connected between the external connection pad 118 and a substrate pad (e.g., the substrate pad 830 to be described later) of the (payment) device (e.g., the smart card) may be secured.

[0052] FIG. 9 is an enlarged plan view (e.g., bottom view) illustrating portion 'Y' in FIG. 4 when the anisotropic conductive layer ACF is compressed. The lower protection layer 520 may further include an accommodation groove 550 in a region other than a region in which the openings 530 in the connection region CR are located. Like the openings 530, the accommodation groove 550 may have a step difference and may provide an accommodation space (corresponding to the accommodation space A of the openings 530) therein. The accommodation groove 550 may be positioned between the openings 530. The accommodation groove 550 may have a rectangular shape but is not limited thereto. The accommodation groove 550 may trap conductive balls CB in

the accommodation space of the accommodation groove **550** so as to prevent the conductive balls CB in the anisotropic conductive layer ACF from escaping the openings **530** and/or the accommodation groove **550** when the anisotropic conductive layer ACF is compressed.

[0053] In example embodiments, the molding member **600** may be provided to be on (e.g., overlap or cover) the control portion **400** including the sensor chip **410** and the passive elements **420** in the mounting region MR on the lower surface **102** of the package substrate **100**. The molding member **600** may serve to protect the sensor chip **410** and the passive elements **420** from external influences such as contamination, impact, and the like. The molding member **600** may include, for example, an epoxy mold compound (EMC). The molding member **600** may include, for example, a UV resin, a polyurethane resin, a silicon resin, a silica filler, and the like.

[0054] The conductive balls CB in the anisotropic conductive layer ACF may be trapped in the openings **530** without escaping the fingerprint sensor package **10** when performing a compression process for mounting the fingerprint sensor package **10** on a (payment) device (e.g., a smart card).

[0055] Thus, short circuits in the fingerprint sensor package **10** due to the escape of the anisotropic conductive layer ACF may be reduced (e.g., prevented), and the conductive balls CB come into contact with the external connection pad **118**, thereby increasing electrical conductivity and improving the stability of bonding between the fingerprint sensor package **10** and a (payment) device (e.g., a smart card).

[0056] Hereinafter, a method of manufacturing a (payment) device (e.g., a smart card) in accordance with example embodiments will be described.

[0057] FIGS. **10** to **18** are views illustrating a method of manufacturing a smart card in accordance with embodiments.

[0058] Referring to FIG. **10**, a base layer **700** on which a fingerprint sensor package **10** will be positioned (e.g., mounted) may be prepared. The base layer **700** may include an open region **710**, an adhesion region **720** extending around (e.g., surrounding) the open region **710** (in a plan view), and a plurality of holes **730** disposed in opposite sides of the open region **710** (in the first direction or in the second direction). For example, in a plan view, the open region **710** may be between a first group of holes among the plurality of holes **730** and a second group of holes among the plurality of holes **730** in the second direction. The fingerprint sensor packages **10** may be disposed in the open regions **710** of the base layer **700**, respectively. A width and a length of the base layer **700** may be determined in accordance with the number and size of the fingerprint sensor packages **10** to be positioned (e.g., mounted) thereon. The base layer **700** may be provided in the form of a reel in which the plurality of holes **730** are configured to be wound or unwound through a winding reel mechanism.

[0059] Referring to FIGS. **11** and **12**, the fingerprint sensor package **10** may be aligned and disposed on the base layer **700**. A lower protection layer **520** of a package substrate **100** of the fingerprint sensor package **10** may be adhered to the adhesion region **720** of the base layer **700**. Then, the package substrate **100** may be cut along the cutting region using a punching apparatus to separate an individual fingerprint sensor package **10** from the base layer **700**. The fingerprint sensor package **10** may have a shape in which each corner is rounded. Therefore, it may be possible to effectively reduce (e.g., prevent) cracks from occurring in the corners in the process of cutting the package substrate **100** through a punching process.

[0060] Then, referring to FIG. **13**, an anisotropic conductive layer ACF may be on (e.g., may overlap with in the third direction or cover) the external connection pad **118**. In some embodiments, the anisotropic conductive layer ACF may be on the lower protection layer **520** on the lower surface **102** of the package substrate **100**. For example, the anisotropic conductive layer ACF may overlap with (e.g., cover) the lower protection layer **520** (in the first direction, the second direction, and/or the third direction). The lower surface **102** of the package substrate **100** in FIG. **13** is illustrated to face upward as the fingerprint sensor package **10** is flipped in FIG. **13**.

[0061] Referring to FIG. **14**, a smart card **800** including a groove region **810**, a card substrate **820**,

a substrate pad **830**, and a security chip **840** may be prepared. The card substrate **820** and the security chip **840** for storing financial information may be disposed in the smart card **800**. The groove region **810** may provide a space for positioning (e.g., mounting) the fingerprint sensor package **10**. The substrate pad **830** may be provided in the groove region **810** to electrically connect the fingerprint sensor package **10**, the card substrate **820**, and the security chip **840**. [0062] Referring to FIG. **15**, the fingerprint sensor package **10** may be positioned (e.g., mounted) in the groove region **810**. External connection pads **118** on the lower surface **102** of the package substrate **100** of the fingerprint sensor package **10** may be electrically connected to the substrate pad **830** of the smart card **800**. The anisotropic conductive layer ACF may be provided between the external connection pads **118** and the substrate pad **830**. The anisotropic conductive layer ACF may include a plurality of conductive balls CB therein. When the fingerprint sensor package **10** is pressed to be positioned (e.g., mounted) on the smart card **800**, the external connection pads **118** of the fingerprint sensor package **10** may be electrically connected to the substrate pad **830** of the smart card **800** through the plurality of conductive balls CB.

[0063] FIGS. **16A** and **16B** are enlarged views illustrating portion 'Z' in FIG. **15**. FIGS. **16A** and **16B** are views showing an anisotropic conductive layer ACF2 (corresponding to the anisotropic conductive layer ACF) when a fingerprint sensor package according to a related art is mounted. Referring to FIG. **16A**, a fingerprint sensor package according to the related art does not have a step difference between an external connection pads **218** (corresponding to the external connection pads **118**) and a lower protection layer **620** (corresponding to the lower protection layer **520**). Thus, referring to FIG. **16B**, when the fingerprint sensor package is compressed, an anisotropic conductive layer ACF **2** may escape through a space between the fingerprint sensor package and the groove region **910** (corresponding to the groove region **810**) of a smart card by vertical pressure to be in contact with a ground pattern **240** (corresponding to the ground pattern **140**) of the fingerprint sensor package, thereby causing an electrical short. FIGS. **17A** and **17B** are views illustrating an anisotropic conductive layer ACF when a fingerprint sensor package **10** in accordance with example embodiments is mounted on the (payment) device (e.g., the smart card **800**). Referring to FIG. **17A**, the openings **530** on the lower protection layer **520** of the fingerprint sensor package **10** may have a step difference from that of the external connection pad **118**. Similarly, a plurality of accommodation grooves **550** have a step difference on the protection layer **500** to provide an accommodation space therein. The anisotropic conductive layer ACF may be guided by the accommodation grooves **550** and/or the openings **530** of the protection layer pattern **540** such that the conductive balls CB are accommodated therein. Thus, as illustrated in FIG. **17B**, the anisotropic conductive layer ACF may remain in the accommodation space A provided between the substrate pad **830** of the smart card **800** and the connection region CR on the lower surface **102** of the package substrate **100** of the fingerprint sensor package **10**. Compared to a fingerprint sensor package according to the related art in FIGS. **16A** and **16B**, more conductive balls CB may stay between the external connection pad **118** and the substrate pad **830**, thereby increasing electrical conductivity and thus the fingerprint sensor package **10** may be stably bonded.

[0064] Referring to FIG. **18**, a smart card **800** may include the fingerprint sensor package **10** and the security chip **840**. The fingerprint of the user may be recognized by contacting the fingerprint sensor package **10** of the smart card **800** with the fingerprint. When the recognized and registered fingerprints match, the security chip **840** may grant the user payment authority.

[0065] Hereinafter, a fingerprint sensor package (corresponding to the fingerprint sensor package **10**) in accordance with example embodiments of the present inventive concepts and a smart card including the fingerprint sensor package will be described.

[0066] FIGS. **19** to **21** are views illustrating a fingerprint sensor package in accordance with example embodiments of the present inventive concepts. FIG. **19** is a cross-sectional view illustrating a fingerprint sensor package **11** in accordance with example embodiments. FIG. **20** is a bottom view illustrating the fingerprint sensor package **11** of FIG. **19**. Components constituting the

fingerprint sensor package **11** described below are substantially the same as or similar to those of the fingerprint sensor package **10** described in FIGS. **1** to **7**, and thus differences will be mainly described.

[0067] Referring to FIGS. **19** and **20**, the fingerprint sensor package **11** may include a first substrate **101**, a second substrate **103**, a sensing portion **200** provided on the first substrate **101**, a coating member **300** on (e.g., overlapping in the third direction or covering) the sensing portion **200**, a control portion **400** configured to process information obtained from the sensing portion **200**, a protection layer **500**, and a molding member **600** on (e.g., overlapping or covering) the control portion **400**.

[0068] The first substrate **101** may include a plurality of first, second, and third insulating layers **110**, **120**, and **130**, a plurality of first and second redistribution wirings **114** and **124** provided in the first, second, and third insulating layers **110**, **120**, and **130**, a plurality of first, second, and third through vias **112**, **122**, and **132**, electrode pads **116**, and first bonding pads **111**. The first substrate **101** may have a rectangular or square shape in a plan view but is not limited thereto.

[0069] An upper surface of the first substrate **101**, that is, an upper surface of the third insulating layer **130** may include a sensing region SR and a substrate adhesion region extending around (e.g., surrounding) the sensing region SR in a plan view. The sensing region SR may be a region in which the sensing portion **200** including a first sensing pattern **210** and a second sensing pattern **220** for fingerprint recognition is disposed. The sensing region SR may be positioned in a central region of the third insulating layer **130**, but is not limited thereto. The sensing region SR may have a square shape in a plan view. The substrate adhesion region may be a region in which an adhesive layer **105** for bonding the first substrate **101** to the second substrate **103** is disposed. The adhesive layer **105** may be disposed (in a band shape in a plan view) on the substrate adhesion region extending around (e.g., surrounding) the sensing region SR in a plan view.

[0070] A lower surface **102** of the first substrate **101**, that is, a lower surface of the first insulating layer **110** may include a mounting region MR and a connection region CR extending around (e.g., surrounding) the mounting region MR. The mounting region MR may be a region in which the control portion **400** including a sensor chip **410** and passive elements **420** is located. The connection region CR may be a region in which the first bonding pads **111** for electrically connecting the first substrate **101** to the second substrate **103** are located. The first bonding pads **111** may be disposed on the first insulating layer **110** to be spaced apart from each other along the periphery of the mounting region MR.

[0071] The second substrate **103** may have an upper surface and a lower surface opposite to the upper surface in the third direction. The second substrate **103** may include an open region OR having a shape corresponding to the shape of the first substrate **101** so as to expose the upper surface of the coating member **300**. The lower surface of the second substrate **103** may include an adhesion region **720**, which overlaps at least a portion of the first substrate **101** when viewed in a plan view. Accordingly, the second substrate **103** may be fixedly connected to the first substrate **101** through the adhesive layer **105** disposed in the adhesion region **720**.

[0072] The second substrate **103** may include a ground pattern **140** on an upper surface thereof. The ground pattern **140** may have a band shape extending around (e.g., surrounding) the open region OR. The ground pattern **140** may be configured to provide a reference potential to the sensor chip **410** and may be configured to serve to shield noise when the sensing portion **200** senses a user's fingerprint.

[0073] As illustrated in FIG. **20**, second bonding pads **113** may be provided on the lower surface of the second substrate **103** along an inner circumference (of the second substrate **103**) adjacent (close to) the first substrate **101**. The second bonding pads **113** may be electrically connected to the first bonding pads **111** of the first substrate **101** using conductive wires **115**. External connection pads **118** may be provided on the lower surface of the second substrate **103** along an outer circumference far from the first substrate **101** (e.g., farther from the first substrate **101** than the second bonding

pads **113**). For example, in a cross-sectional view, the external connection pads **118** may be disposed farther from the second bonding pads **113** in the first direction and/or the second direction. The external connection pads **118** may be electrically connected to substrate pads **830** when the fingerprint sensor package **11** is mounted on the smart card **800**.

[0074] A lower protection layer **520** may be provided on the lower surface of the second substrate **103**. The lower protection layer **520** may cover the lower surface of the second substrate **103** to expose (at least a portion of) the second bonding pads **113**. The lower protection layer **520** may include openings **530** that expose (at least a portion of) the external connection pads **118**. The openings **530** have a step difference from the external connection pads **118** and may provide an accommodation space A therein. In embodiments, the lower protection layer **520** may further include a protection layer pattern **540** and an accommodation groove **550**.

[0075] The molding member **600** may be provided to be on (e.g., overlap or cover) the control portion **400** including the sensor chip **410** and the passive elements **420** on the lower surface **102** of the first substrate **101**, the first bonding pads **111**, the second bonding pads **113** on the lower surface of the second substrate **103**, and the conductive wires **115** (electrically) connecting the first bonding pads **111** to the second bonding pads **113**. The molding member **600** may include, for example, an epoxy mold compound (EMC). The molding member **600** may include, for example, a UV resin, a polyurethane resin, a silicon resin, a silica filler, etc.

[0076] Referring to FIG. **21**, a smart card **800** including a groove region **810**, a card substrate **820**, a substrate pad **830**, and a security chip **840** may be prepared. The fingerprint sensor package **11** may be mounted in the groove region **810**. The external connection pads **118** on the lower surface of the second substrate **103** of the fingerprint sensor package **11** may be electrically connected to the substrate pads **830** of the smart card **800**. An anisotropic conductive layer ACF may be provided between the external connection pads **118** and the substrate pads **830**. The anisotropic conductive layer ACF may include a plurality of conductive balls CB therein. When the fingerprint sensor package **11** is compressed to be mounted on the smart card **800**, the external connection pads **118** of the fingerprint sensor package **11** may be electrically connected to the substrate pad **830** of the smart card **800** through the plurality of conductive balls CB.

[0077] The foregoing is illustrative of example embodiments and is not to be construed as limiting thereof. Although a few example embodiments have been described, those skilled in the art will readily appreciate that various modifications are possible in some example embodiments without materially departing from the scope, novel teachings, and advantages of the present invention. Accordingly, all such modifications are intended to be included within the scope of example embodiments as defined in the claims.

Claims

1. A fingerprint sensor package, comprising: a package substrate including an upper surface and a lower surface that is opposite to the upper surface in a first direction, wherein the first direction is perpendicular to the upper surface of the package substrate, wherein the upper surface of the package substrate includes a sensing region and a peripheral region that extends around the sensing region, and wherein the lower surface of the package substrate includes a mounting region and a connection region that extends around the mounting region; external connection pads in the connection region on the lower surface of the package substrate; a plurality of capacitor patterns in the sensing region on the upper surface of the package substrate; a control portion in the mounting region on the lower surface of the package substrate, wherein the control portion includes a sensor chip and a passive element that is adjacent the sensor chip; a lower protection layer on the lower surface of the package substrate, wherein the lower protection layer includes openings that overlap at least a portion of the external connection pads in the first direction respectively, wherein the lower protection layer includes a protection layer pattern that overlaps the external connection pads

in the first direction in the openings; and an anisotropic conductive layer on the external connection pads in the openings and on the lower protection layer.

2. The fingerprint sensor package of claim 1, wherein the protection layer pattern comprises: a plurality of first extension portions extending between a first side portion and a third side portion of each of the openings, wherein the third side portion faces the first side portion in a second direction; and a plurality of second extension portions extending between a second side portion and a fourth side portion of each of the openings, wherein the fourth side portion faces the second side portion in a third direction, wherein the first extension portions and the second extension portions intersect each other to provide a plurality of accommodation cavities that overlap at least a portion of the external connection pads in the first direction, wherein the second direction and the third direction are parallel with the upper surface of the package substrate, and wherein the second direction and the third direction intersect with each other.

3. The fingerprint sensor package of claim 2, wherein each of the accommodation cavities has a rectangular or circular shape in a plan view.

4. The fingerprint sensor package of claim 1, wherein the protection layer pattern comprises: a plurality of first protrusion portions protruding from a first side portion of each of the openings to extend in a second direction; and a plurality of second protrusion portions protruding from a second side portion of each of the openings to extend in the second direction, wherein the second side portion faces the first side portion in the second direction, wherein the first side portion and the second side portion extend in a third direction, wherein the first protrusion portions are spaced apart from each other in the third direction, wherein the second protrusion portions are spaced apart from each other in the third direction, wherein the second direction and the third direction are parallel with the upper surface of the package substrate, and wherein the second direction and the third direction are perpendicular to each other.

5. The fingerprint sensor package of claim 1, wherein an area of a portion of the external connection pads that is overlapped by the protection layer pattern is less than 20% of an area of the external connection pads.

6. The fingerprint sensor package of claim 1, wherein the openings have a step difference from the external connection pads in the first direction to have an accommodation space, and the step difference is within a range of 15 μm to 25 μm .

7. The fingerprint sensor package of claim 1, wherein the lower protection layer further comprises accommodation grooves between the openings, each of the accommodation grooves having an accommodation space therein.

8. The fingerprint sensor package of claim 7, wherein each of the accommodation grooves has a rectangular or circular shape in a plan view.

9. The fingerprint sensor package of claim 1, further comprising: a molding member on the sensor chip and the passive element.

10. A fingerprint sensor package, comprising: a package substrate including: a first substrate having a sensing region and an adhesion region extending around the sensing region on an upper surface thereof and having first bonding pads on a lower surface thereof; and a second substrate having second bonding pads and external connection pads on a lower surface thereof, the second bonding pads being electrically connected to the first bonding pads through conductive wires, the second substrate having an opening region that overlaps the sensing region of the upper surface of the first substrate in a first direction, the second substrate being adhered on the adhesion region of the upper surface of the first substrate; a plurality of capacitor patterns in the sensing region on the upper surface of the first substrate; a control portion on the lower surface of the first substrate, the control portion including a sensor chip and a passive element adjacent the sensor chip; a lower protection layer on the lower surface of the second substrate and having openings that overlap the external connection pads in the first direction respectively, the lower protection layer having a protection layer pattern that overlaps a portion of the external connection pads in the openings in the first

direction; and an anisotropic conductive layer on the external connection pads in the openings and on the lower protection layer, wherein the first direction is perpendicular to the upper surface of the first substrate.

11. The fingerprint sensor package of claim 10, wherein the protection layer pattern comprises: a plurality of first extension portions extending between a first side portion and a third side portion of each of the openings, wherein the third side portion faces the first side portion in a second direction; and a plurality of second extension portions between from a second side portion and a fourth side portion of each of the openings, wherein the fourth side portion faces the second side portion in a third direction, wherein the first extension portions and the second extension portions intersect each other to provide a plurality of accommodation cavities that overlap at least a portion of the external connection pads in the first direction, wherein the second direction and the third direction are parallel with the upper surface of the first substrate, and wherein the second direction and the third direction intersect with each other.

12. The fingerprint sensor package of claim 11, wherein each of the accommodation cavities has a rectangular or circular shape in a plan view, and wherein the anisotropic conductive layer is in a band shape in a plan view.

13. The fingerprint sensor package of claim 10, wherein the protection layer pattern comprises: a plurality of first protrusion portions protruding from a first side portion of each of the openings to extend in a second direction; and a plurality of second protrusion portions protruding from a second side portion of each of the openings to extend in the second direction, wherein the second side portion faces the first side portion in the second direction, wherein the first side portion and the second side portion extend in a third direction, wherein the first protrusion portions are spaced apart from each other in the third direction, wherein the second protrusion portions are spaced apart from each other in the third direction, wherein the second direction and the third direction are parallel with the upper surface of the first substrate, and wherein the second direction and the third direction are perpendicular to each other.

14. The fingerprint sensor package of claim 10, wherein an area of a portion of the external connection pads that is overlapped by the protection layer pattern is less than 20% of an area of the external connection pads.

15. The fingerprint sensor package of claim 10, wherein the openings have a step difference from the external connection pads in the first direction to have an accommodation space, and the step difference is within a range of 15 μm to 25 μm .

16. The fingerprint sensor package of claim 10, wherein the lower protection layer further comprises accommodation grooves between the openings, each of the accommodation grooves having an accommodation space therein.

17. The fingerprint sensor package of claim 16, wherein each of the accommodation grooves has a rectangular or circular shape in a plan view.

18. A fingerprint sensor package, comprising: a package substrate including an upper surface and a lower surface that is opposite to the upper surface in a first direction, wherein the first direction is perpendicular to the upper surface of the package substrate, wherein the upper surface of the package substrate includes a first region and a second region that extends around the first region, and wherein the lower surface of the package substrate includes a third region and a fourth region that extends around the third region; external connection pads in the fourth region on the lower surface of the package substrate; a plurality of first sensing patterns in the first region, extending in a second direction, being spaced apart from each other in a third direction; a plurality of second sensing patterns in the first region, wherein the plurality of second sensing patterns are spaced apart from the plurality of first sensing patterns in the first direction, and wherein the plurality of second sensing patterns extend in the third direction and spaced apart from each other in the second direction; a control portion in the third region, the control portion including a sensor chip and a passive element adjacent the sensor chip; a lower protection layer on the lower surface of the

package substrate and having openings that overlap at least a portion of the external connection pads in the first direction respectively, and the lower protection layer including a protection layer pattern that overlaps the external connection pads in the first direction in the openings; and an anisotropic conductive layer on the external connection pads in the openings and on the lower protection layer, wherein the second direction and the third direction are parallel with the upper surface of the package substrate, and wherein the second direction and the third direction intersect with each other.

19. The fingerprint sensor package of claim 18, wherein the protection layer pattern comprises: a plurality of first extension portions extending between a first side portion and a third side portion of each of the openings, wherein the third side portion faces the first side portion in the second direction; and a plurality of second extension portions extending between a second side portion and a fourth side portion, wherein the fourth side portion faces the second side portion in the third direction, and wherein the first extension portions and the second extension portions intersect each other to provide a plurality of accommodation cavities that overlap at least a portion of the external connection pads in the first direction.

20. The fingerprint sensor package of claim 18, wherein the protection layer pattern comprises: a plurality of first protrusion portions protruding from a first side portion of each of the openings to extend in the second direction; and a plurality of second protrusion portions protruding from a second side portion of each of the openings to extend in the second direction, wherein the second side portion faces the first side portion in the second direction, wherein the first side portion and the second side portion extend in the third direction, wherein the first protrusion portions are spaced apart from each other in the third direction, and wherein the second protrusion portions are spaced apart from each other in the third direction.
